CRM Analytics Dashboard

Power BI Project Documentation

Internship Project Work

Project	End-to-End CRM Analytics Dashboard in Power BI
Tool	Microsoft Power BI Desktop
Data Source	Multi-table Synthetic CRM Dataset (Excel)
Deliverable	5 Interactive Dashboard Pages + Documentation
Role	Intern – Dashboard Development & Analytics

This document describes the complete CRM analytics dashboard developed during the internship.

All data used is synthetic / demo data created for project purposes.

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1. Project Summary

As part of my internship, I designed and developed a complete CRM Analytics Dashboard using Microsoft Power BI. The goal was to create an executive and operational reporting solution that gives clear visibility into the full customer lifecycle — from lead generation to revenue.

I worked with a multi-table synthetic CRM dataset and built five interactive pages covering leads, AI & call performance, follow-ups, appointments, revenue and customer health. The dashboard uses a consistent design language (dark sidebar navigation, KPI cards, modern charts) and supports filtering, cross-filtering and reset functionality.

What I Delivered

- 5 fully interactive Power BI report pages with unified navigation
- End-to-end metrics from Lead → Qualification → Follow-up → Appointment → Conversion → Revenue
- Dedicated AI performance analytics (interactions, resolution rate, response time, handoffs)
- Actionable tables for high-priority and overdue follow-ups
- Clean, professional visual design suitable for management presentation
- This documentation file for future reference and handover

2. Objectives & Scope of Work

Objectives of this task

- Build a single Power BI solution that covers the complete CRM funnel.
- Show lead quality and pipeline health, not just volume.
- Measure AI automation effectiveness versus human involvement.
- Track follow-up discipline and appointment conversion.
- Present revenue and customer health in a clear, actionable way.
- Maintain consistent UI/UX across all pages.

Scope Boundaries

- Static Excel dataset (no live CRM connection in this version).
- No row-level security (RLS) implemented.
- No predictive / forecasting models.
- Desktop-first layout (mobile optimization left for later).

3. Dataset Used

The dashboard is powered by a multi-table synthetic Excel file created for this project. All records are fictional and intended only for demonstration and learning purposes.

Tables in the Dataset

Table	Approx. Rows	What it contains
Clients	~85	Client master (Industry, City, Status, Acquisition source)
Leads	~1,250	Lead records with source, status, score, temperature
AI_Interactions	~3,240	AI interactions, response time, outcomes, handoffs
Calls	~1,860	Call logs, outcome, agent type (AI / Human)
Followups	~1,700	Follow-up activities, status, priority, channel
Appointments	~400	Appointments, status, attendance, conversion after meeting
Customers	~167	Converted customers, status, revenue, renewal info
Revenue	~340	Revenue transactions, payment status, recurring flag

Relationships were created mainly on Client_ID, Lead_ID and Customer_ID so that metrics can be analysed across tables.

4. Dashboard Structure

I designed a consistent layout across all pages: dark navy left sidebar for navigation, white main canvas, uniform KPI cards, and shared date + filter + reset controls on the top-right.

Sr. No.	Page	Focus of the page
1	Executive Overview	High-level business health – leads, conversion, revenue, operational KPIs
2	Lead Performance	Lead quality, score, temperature, source quality, pipeline movement
3	AI & Call Performance	AI interactions, resolution, response time, handoffs, call outcomes
4	Follow-up & Appointments	Follow-up discipline, overdue items, appointment status & conversion
5	Revenue & Customers	Revenue trends, payment status, customer health, top customers

5. Page-wise Details

5.1 Executive Overview

Landing page that gives a quick snapshot of overall performance. Main question answered: “How is the business performing right now?”

What I included:

- KPI cards for Total Leads, Qualified Leads, Conversion Rate, Total Customers, Total Revenue, AI Interactions, Pending Leads/Follow-ups, Appointments, Human Handoffs
- Charts: Leads Over Time, Leads by Source, Leads by Status, Industry Performance, Conversion Funnel, Revenue Over Time

5.2 Lead Performance

Focused on quality of leads and pipeline health instead of only volume. Main question: “How healthy is our lead pipeline and where are we losing leads?”

What I included:

- Quality KPIs: Average Lead Score, Qualification Rate, Hot Lead Rate, Contact / Proposal / Lost rates
- Lead Score Distribution, Temperature vs Conversion, Lead Pipeline Funnel
- Matrices for Source Quality and Industry Quality
- Pending Lead Aging + High-Priority Leads table (actionable)

5.3 AI & Call Performance

Measures how well AI and human communication channels are performing. Main question: “Is AI resolving queries well and where do we still need humans?”

What I included:

- KPIs: AI Interactions, Total Calls, Human Handoffs, AI Resolved, Resolution Rate, Avg Response Time
- Trends for Interactions, Resolution Rate and Response Time
- AI vs Human view, Call Outcome Distribution, Performance by Channel
- Interaction Types, Top Query Categories, Handoff by Industry

5.4 Follow-up & Appointments

Tracks whether follow-ups are done on time and how well appointments convert. Main question: “Are we following up properly and converting meetings?”

What I included:

- KPIs: Total / Pending / Overdue Follow-ups, Appointments Booked & Completed, Show Rate
- Follow-ups Over Time, Status distributions, Priority & Type breakdowns
- Appointment Conversion Funnel
- Industry performance matrix + Overdue & High-Priority Follow-ups table

5.5 Revenue & Customers

Financial and customer health view. Main question: "Where is revenue coming from and how healthy are our customers?"

What I included:

- KPIs: Total Revenue, Paid Revenue, Total & Active Customers, ARPC, Recurring Revenue %
- Revenue Over Time (combo), Payment Status & Transaction Type breakdowns
- Revenue by Industry & Service, Customer Status matrix
- Top Customers table, Pricing Plan view, Gauge for Recurring Revenue %

6. Key DAX Measures Implemented

I created a set of reusable DAX measures that power the cards and charts across all pages. Some of the most important ones are listed below.

Lead related

```
Total Leads = DISTINCTCOUNT(Leads[Lead_ID])
Qualified Leads = CALCULATE(DISTINCTCOUNT(Leads[Lead_ID]), Leads[Lead_Status] = "Qualified")
Converted Leads = CALCULATE(DISTINCTCOUNT(Leads[Lead_ID]), Leads[Converted] = "Yes")
Conversion Rate = DIVIDE([Converted Leads], [Total Leads], 0)
Average Lead Score = AVERAGE(Leads[Lead_Score])
```

AI & Calls related

```
AI Interactions = DISTINCTCOUNT(AI_Interactions[Interaction_ID])
AI Resolved = CALCULATE(..., Interaction_Outcome = "Resolved")
AI Resolution Rate = DIVIDE([AI Resolved], [AI Interactions], 0)
Avg AI Response Time = AVERAGE(AI_Interactions[Response_Time_Seconds])
Human Handoffs = CALCULATE(..., Human_Handoff = "Yes")
```

Follow-up, Appointment & Revenue related

```
Total / Pending / Overdue Follow-ups (using Followup_Status)
Appointment Show Rate = DIVIDE([Completed Appointments], [Appointments Booked], 0)
Total Revenue = SUM(Revenue[Amount])
Paid Revenue = CALCULATE(SUM(Revenue[Amount]), Payment_Status = "Paid")
Recurring Revenue % = DIVIDE([Recurring Revenue], [Total Revenue], 0)
Average Revenue per Customer = DIVIDE([Total Revenue], [Total Customers], 0)
```

7. Filters, Slicers & Interactivity

I kept the filter experience the same on every page so that users do not have to re-learn controls.

Controls available on every page

- Date Range Slicer – filters the page by relevant date columns
- Filters button – additional slicers (Industry, Source, Status, etc.)
- Reset (circular) button – Clear all slicers action so users can quickly go back to the full view

Other interactivity

- Clicking any chart segment cross-filters related visuals on the same page.
- Left sidebar is used for page navigation; the active page is highlighted.

8. How to Open and Use

Requirements

- Microsoft Power BI Desktop (recent version recommended)
- The .pbix file of the dashboard

Steps to open

- Open Power BI Desktop
- File → Open → select the .pbix file
- All five pages load and are ready for interaction

Refreshing data (if needed)

- Point the data source to the correct Excel file location if the path has changed
- Home → Refresh

Publishing (optional)

- Home → Publish → select a workspace
- Open the report in Power BI Service for browser viewing and sharing
- Note: A Microsoft work/school account is usually required for publishing.

9. Technical Implementation Notes

- Data model: Relationships created mainly on Client_ID, Lead_ID and Customer_ID.
- Date handling: Native date hierarchies used for monthly trends.
- Calculated columns: Lead Score Band and Pending Age Band created for distribution and aging analysis.
- Disconnected tables: Used for custom funnel stage order (Pipeline / Appointment funnels).
- Visuals used: Card, Line, Column, Bar, Donut, Funnel, Matrix, Table, Gauge.
- Design system: Consistent navy sidebar, white cards, light borders, soft shadows, blue accent.

10. Assumptions & Limitations

Assumptions

- Dataset is fully synthetic and created only for this project / demonstration.
- Conversion is primarily identified via the Converted flag in the Leads table.
- AI “Resolved” status is taken from Interaction_Outcome = "Resolved".

Limitations of the current version

- Static data – no live connection to a production system.
- No row-level security.
- Mobile layout not optimised.
- Some advanced features (custom tooltips, complex bookmarks) kept simple for clarity.

11. Final Insights and Recommendations

Based on the analysis enabled by this dashboard and the structure of the underlying data, the following insights and practical recommendations can be drawn.

Key Insights

- Lead quality varies significantly by source. Volume alone is not a reliable indicator of performance — conversion rate and average lead score by source give a clearer picture.
- A large share of pending leads sit in older age buckets (especially 30+ days). This indicates follow-up discipline needs tighter monitoring.
- AI handles a substantial portion of interactions, but human handoffs remain important for complex or high-value cases. Resolution rate and response time trends help track AI effectiveness over time.
- Appointment show rate and post-appointment conversion are critical middle-of-funnel metrics. Improving confirmation and reminder processes can lift overall conversion.
- Recurring revenue forms a major part of total revenue. Tracking recurring vs one-time revenue and customer status (Active / Renewal Due / Inactive) supports retention focus.
- Industry-level differences exist in lead quality, handoff rates and revenue contribution. Targeting and playbooks can be refined by industry.

Recommendations

- Prioritise High and Overdue follow-ups daily using the actionable table on the Follow-up page. Set internal SLAs for first response and subsequent follow-ups.
- Review low-converting lead sources and either improve messaging/qualification or reallocate effort toward higher-quality channels.
- Use AI Resolution Rate and Average Response Time as ongoing KPIs. Investigate drops in resolution or spikes in response time promptly.
- Strengthen appointment confirmation and reminder workflows to improve show rate and downstream conversion.
- Monitor customers with Renewal Due status and design proactive outreach to reduce churn.
- Extend the dashboard later with live data connection, row-level security (if multi-user), and optional forecasting once historical patterns stabilise.
- Keep the current five-page structure as the core; add industry or team deep-dive pages only when there is a clear operational need.

How the dashboard supports these actions

The Executive Overview gives leadership a quick health check. Lead Performance and Follow-up pages surface where pipeline and process issues sit. AI & Call Performance quantifies automation impact. Revenue & Customers links operational activity to financial outcomes. Together, the pages move the team from “what happened” to “what should we do next.”

12. Skills Demonstrated & Conclusion

Skills and practices applied in this task

- Power BI report design and multi-page architecture
- Data modelling and relationships across multiple tables
- DAX measure writing for KPIs, rates and ratios
- Calculated columns for banding and aging analysis
- Visual selection and storytelling (funnels, quality matrices, actionable tables)
- Consistent UI/UX and navigation design
- Documentation and handover readiness

Conclusion

Through this internship task I delivered a complete, presentation-ready CRM analytics dashboard. The five pages together cover the full journey from lead acquisition to revenue and give both executive and operational users a clear, interactive view of performance. The solution is structured so that it can be extended later with live data sources, additional pages or more advanced analytics without redesigning the existing foundation.

— End of Documentation —

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